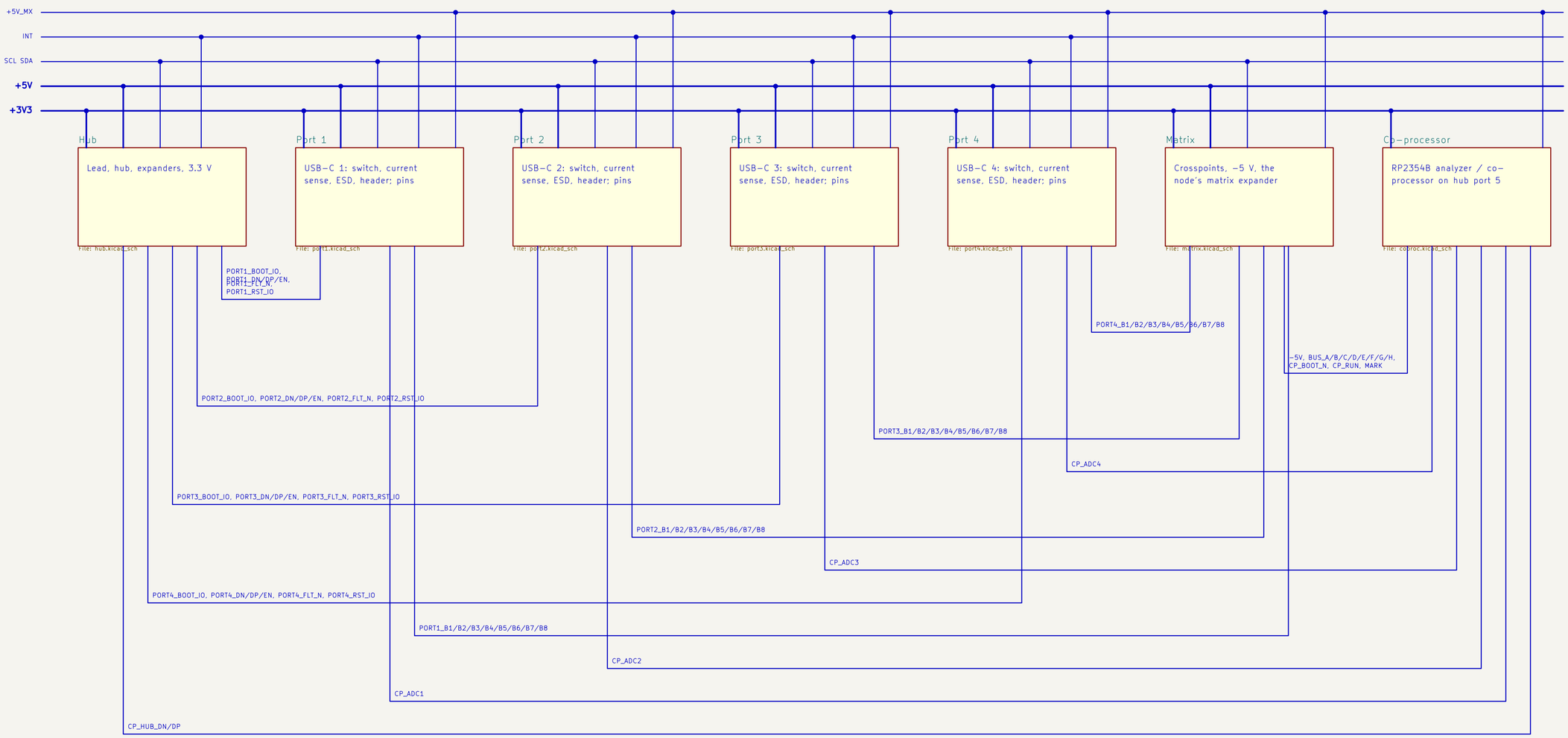


DATA ×4 v2.0 — USB2517I hub (7 ports), four USB-C (sub-slots A1,1-A1,4) each with a TPS2553-1, an INA226 and ESD, two TCA6408A for EN/FAULT/RESET/BOOT, an AP2114H 3.3 V; the PH 8 lead from the node's J7.
Pins: 32 on two CH446Q (±5 V). LSF0108 per port, an RP2354B analyzer on hub port 5, a TCA9534A for the matrix and the co-processor's RUN/BOOTSEL. Standing behind the port wall, one face assembled: v1 stays the other bays' board.
Lead pinout here: 1 INT · 2 SCL · 3 SDA · 4 GND · 5 D− · 6 D+ · 7 GND · 8 +5V (the node's J7 mirrored).
I²C: TCA6408A 0x20 / 0x21, INA226 0x41 / 0x43 / 0x44 / 0x45, TCA9534A 0x38; the node's INA226 0x40.
Assembled as DATA-4x-logic (as drawn) or DATA-4x (BOM/CPL '-base': the matrix, the co-processor and the ports' pin chains not fitted).

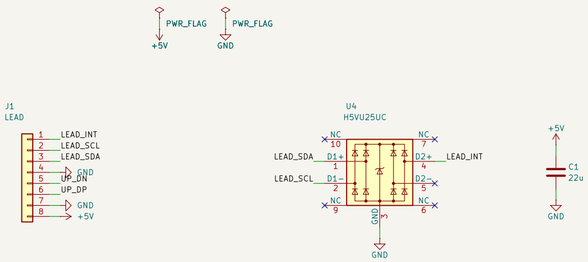
Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack

Sheet: /
File: data-port-x4-v2.kicad_sch

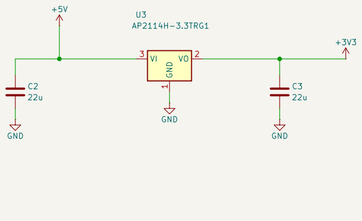
Title: DATA ×4 v2 — Overview

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 1/8

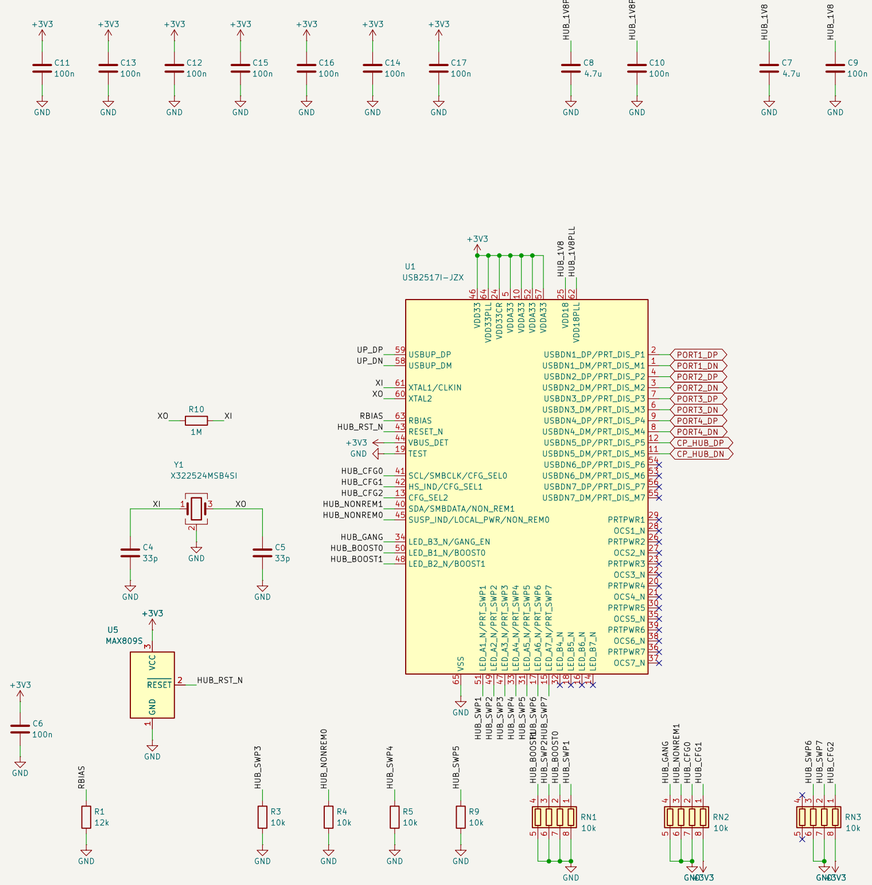
Lead from the bay node's J7 (JST PH 8, vertical, on the rear face): SDA, SCL and INT through a HSVU25UC



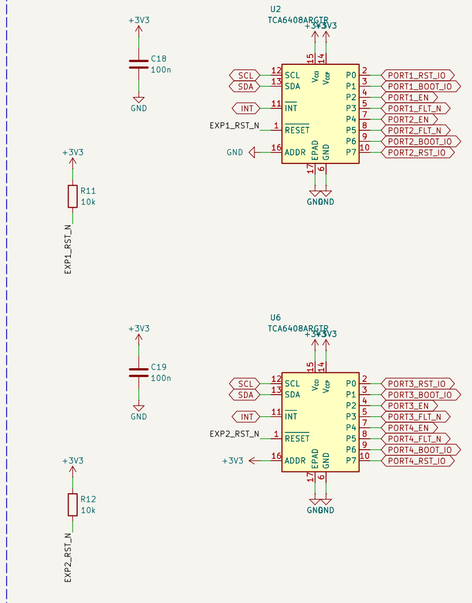
3.3 V for the hub, the expanders and the INA226s (AP2114H, SOT-223 on copper)



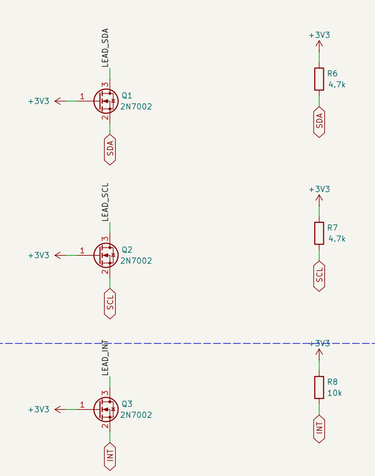
USB2517I hub, CFG_SEL 110 (internal default configuration, straps disabled), 24 MHz crystal



Two TCA6408A expanders, one per pair of ports: EN, FAULT_N, RESET, BOOT



I²C isolation: open with DUT_5V off, so the node's pull-ups cannot feed this board



Hub reset: U5 (MAX809S, 2.93 V, ≥ 150 ms, push-pull) on RESET_N (errata DS80000628A, Module 2).
Hub: CFG_SEL[2:0] = 110 (internal default configuration, strap options disabled: DS00001598C Table 5-2):
self-powered, ports switched and sensed one by one (here by the expanders), no port disabled, D+ / D- not sampled at reset; the other strap pins keep their pull-downs, unread. Ports 6 and 7 enabled, unused (cosmetic). On purpose: PRTPWR unused and OCS_N on their pull-ups (the node switches each port and reads its FAULT_N through the expanders); VBUS_DET on 3.3 V (self-powered hub).
Hub port n = USB-C n (busid 1-1.n) for n = 1-4; port 5 = the co-processor.
Expander pins are inputs at power-up: the EN pull-downs keep every port off until the node writes.
INT (J1.1 → the node's IO4): both expanders' INT and the four INA226 ALERTs, open drain, wired-OR, through Q3.
I²C and INT: the node's pull-ups on LEAD_*, 4.7k / 4.7k / 10k here; with DUT_5V off the chips here lose their registers; the node sets them up again after each DUT_5V on, and leaves this board's addresses alone while off.
J1 (through-hole) stands on the rear face: soldered by hand (the PCBA places the front face only).

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

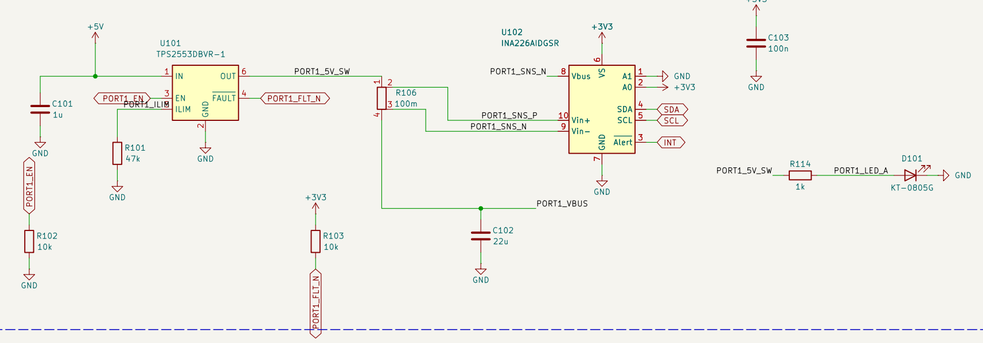
hw-testing-rack

Sheet: /Hub/
File: hub.kicad_sch

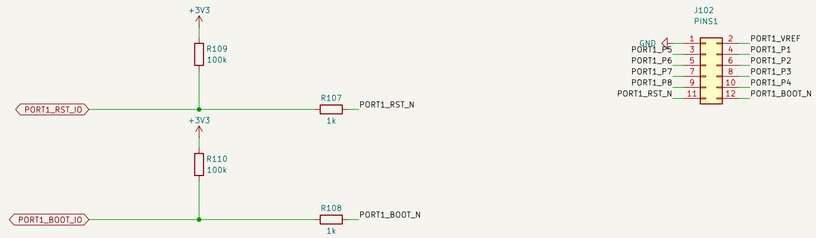
Title: DATA x4 v2 — Lead, hub, expanders, 3.3 V

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 2/8

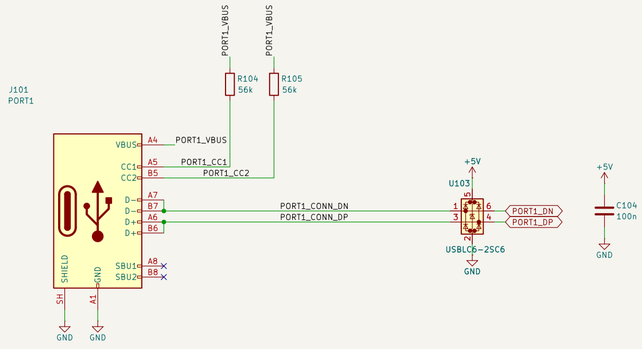
Port 1 power: +5V → TPS2553-1 → PORT1_5V_SW → 100 mΩ shunt → PORT1_VBUS, the INA226 across the shunt



Header 1 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

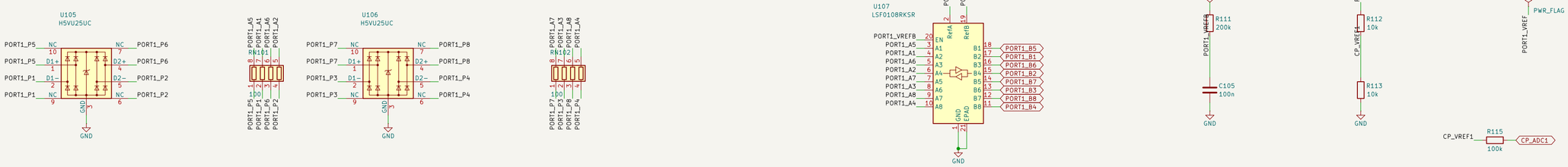


USB-C 1 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)

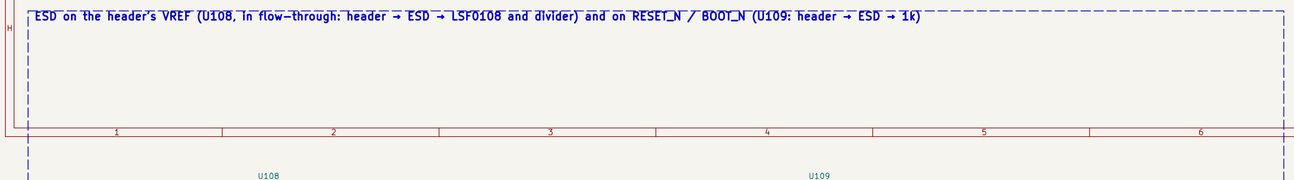


Port 1 = sub-slot A1.1, hub port 1, busid 1-1.1.
INA226 at 0x41: expander U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
  an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's
  crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U108, RESET_N / BOOT_N through U109 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

Pins 1: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines



ESD on the header's VREF (U108, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U109: header → ESD → 1k)



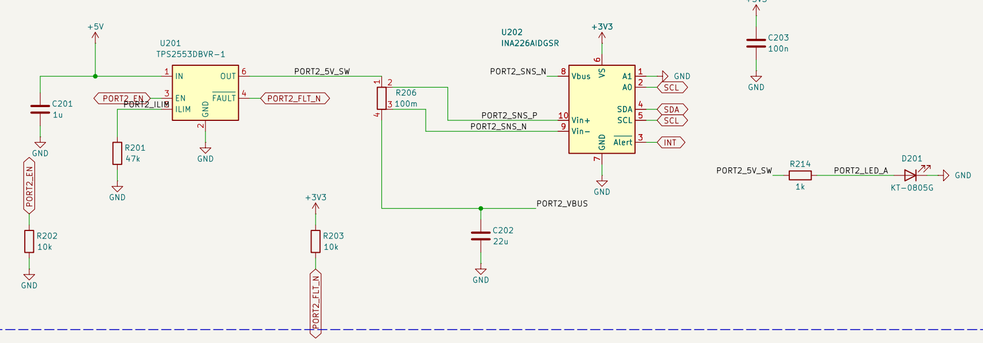
Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack

Sheet: /Port 1/
File: port1.kicad_sch

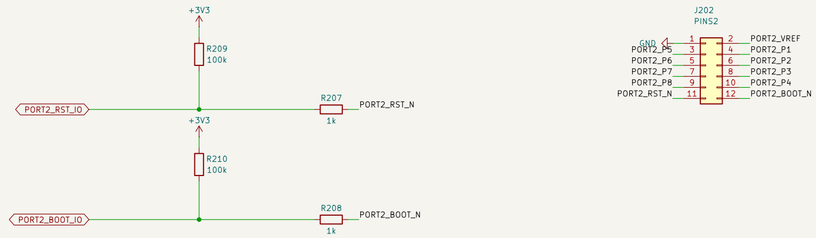
Title: DATA ×4 v2 — USB-C 1: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 3/8

Port 2 power: +5V → TPS2553-1 → PORT2_5V_SW → 100 mΩ shunt → PORT2_VBUS, the INA226 across the shunt

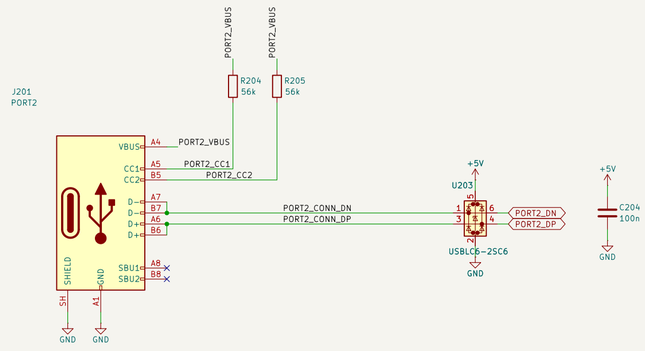


Header 2 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up



Port 2 = sub-slot A1.2, hub port 2, busid 1-1.2.
INA226 at 0x43; expander U2 (0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
  an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's
  crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U208, RESET_N / BOOT_N through U209 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

USB-C 2 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 2: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines



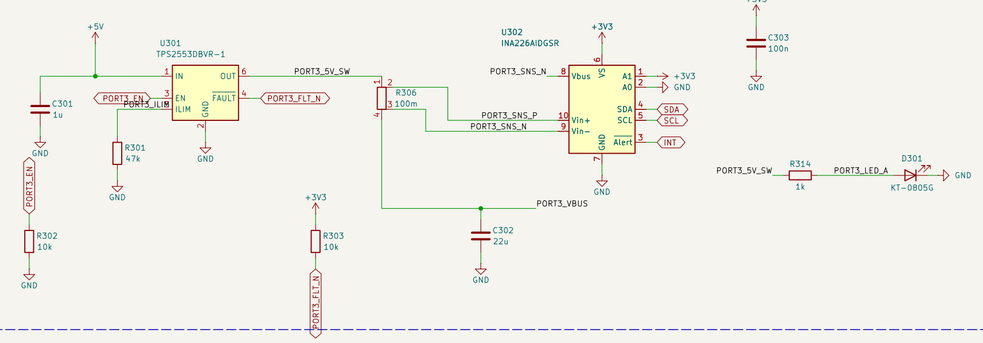
ESD on the header's VREF (U208, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U209: header → ESD → 1k)

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Port 2/
File: port2.kicad_sch

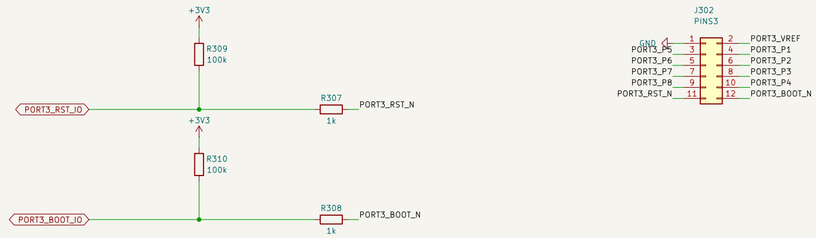
Title: DATA ×4 v2 — USB-C 2: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 4/8

Port 3 power: +5V → TPS2553-1 → PORT3_5V_SW → 100 mΩ shunt → PORT3_VBUS, the INA226 across the shunt

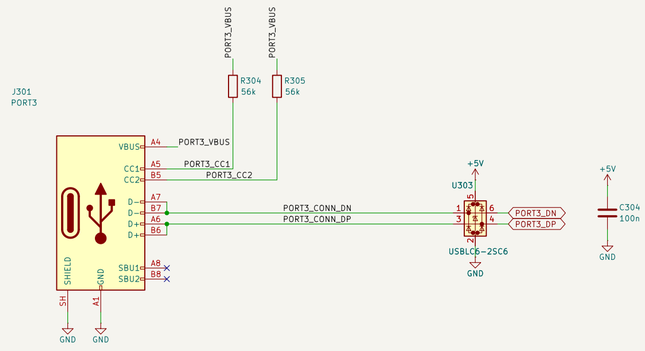


Header 3 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up

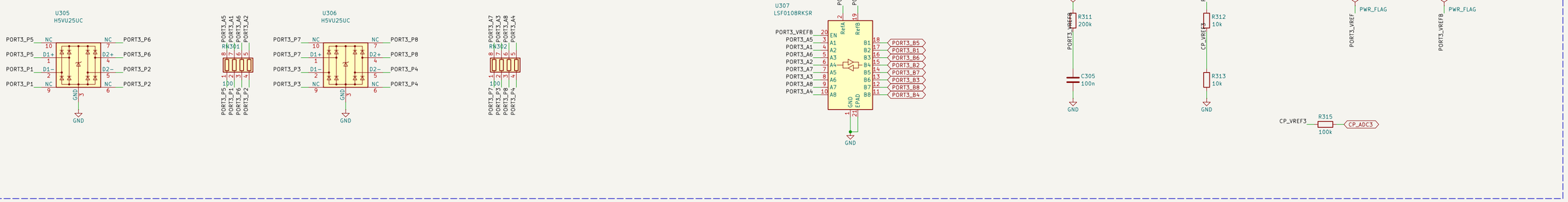


Port 3 = sub-slot A1.3, hub port 3, busid 1-1.3.
INA226 at 0x44: expander U6 (0x21).
ILIM 47k: 0.50-0.62 A; the −1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U308, RESET_N / BOOT_N through U309 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

USB-C 3 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 3: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



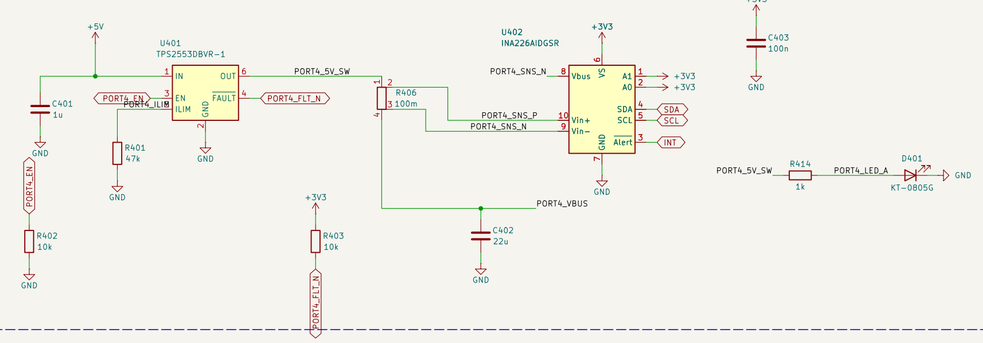
ESD on the header's VREF (U308, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U309: header → ESD → 1k)

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Port 3/
File: port3.kicad_sch

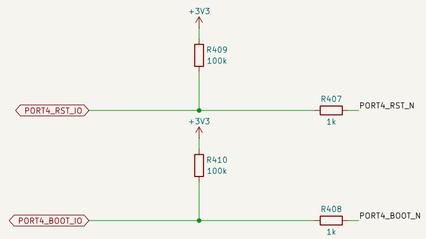
Title: DATA ×4 v2 — USB-C 3: switch, current sense, ESD, header; pins

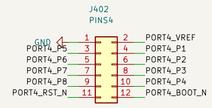
Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 5/8

Port 4 power: +5V → TPS2553-1 → PORT4_5V_SW → 100 mΩ shunt → PORT4_VBUS, the INA226 across the shunt



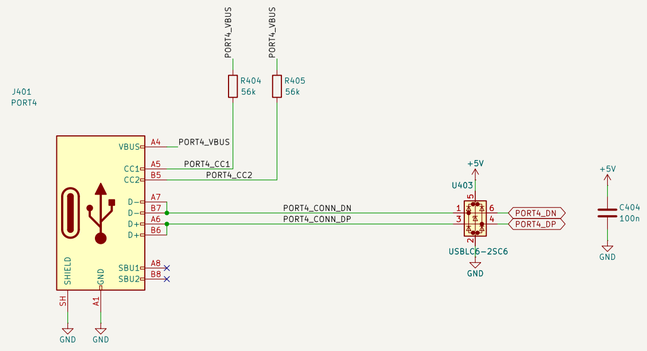
Header 4 (2 × 6 through-hole): GND, VREF, P1-P8, RESET_N, BOOT_N; RESET/BOOT through 1k, their expander lines pulled up



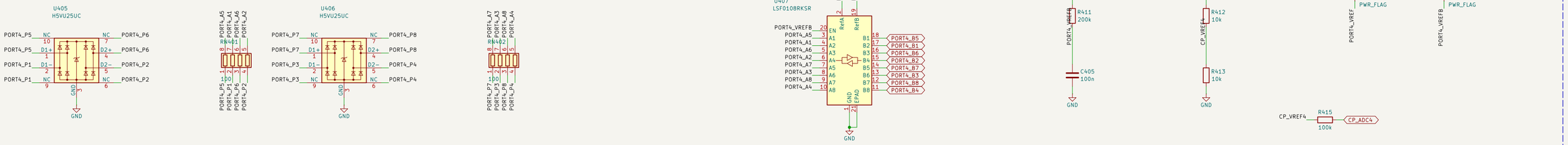


Port 4 = sub-slot A1.4, hub port 4, busid 1-1.4.
INA226 at 0x45; expander U6 (0x21).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → crosspoint X lines (docs/electronics.md › DATA ×4 v2 map):
  an unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the firmware keeps the port's
  crosspoint switches open while its VREF reads under 0.65 V (GPIO40-43 under 0.33 V).
ESD: VREF through U408, RESET_N / BOOT_N through U409 (flow-through).
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
VBUS LED: on while the port's switch is on (5V_SW, before the shunt).

USB-C 4 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Pins 4: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



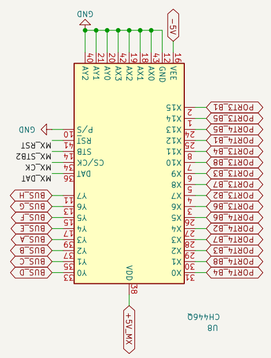
ESD on the header's VREF (U408, in flow-through: header → ESD → LSF0108 and divider) and on RESET_N / BOOT_N (U409: header → ESD → 1k)

Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Port 4/
File: port4.kicad_sch

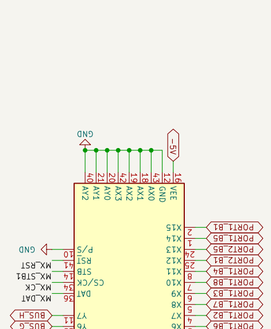
Title: DATA ×4 v2 — USB-C 4: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 2.0**
KiCad E.D.A. 10.0.6		Id: 6/8

Crosspoint U8 (CH446Q at ±5 V, serial addressing): X = ports 3-4's pins, Y0-Y7 = buses DCBAEFGH



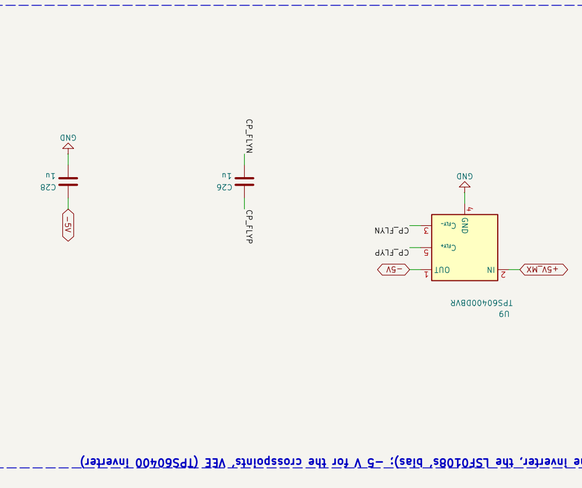
Crosspoint U7 (CH446Q at ±5 V, serial addressing): X = ports 1-2's pins, Y0-Y7 = buses DCBAEFGH



The node's matrix expander (TCA9534, 0x38): the crosspoints and the co-processor's RUN / BOOTSEL

CH446Q serial mode (P/S low): 7 address bits on CK, data on DAT, latched by STB; RST high = all open.
TCA9534 0x38: P0 DAT, P1 CK, P2 STB, P3 STB, U8, P4 RST, P5 RUN, P6 BOOTSEL, P7 mark.
RUN and BOOTSEL are open drain emulated (output register 0, the pin switched to output).
±5 V on purpose: at a single 5 V R_on is 120-225 Ω, and a 1.8 V board's I²C low would fail.

+5V_MX through a ferrite (the crosspoints, the inverter, the LSF0108' bias): -5 V for the crosspoints' VEE (TPS60400 inverter)

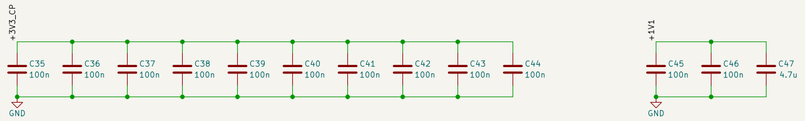


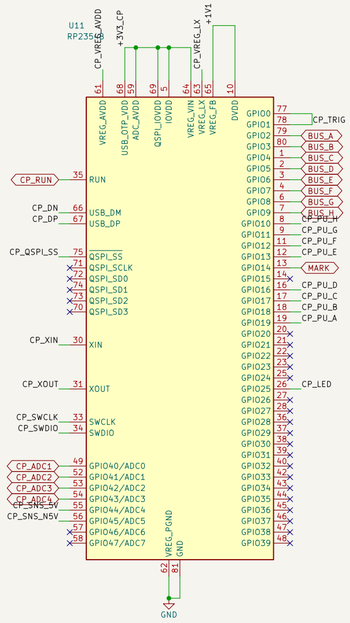
Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an

Title: DATA ×4 v2 — Crosspoints, −5 V, the node's matrix expander

Sheet: /Matrix/	
File: matrix.kicad_sch	
Size: A2	Date:
KiCad E.D.A. 10.0.6	Id: 7/8
hw-testing-rack	**Rev: 2.0**

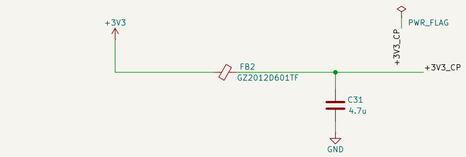
RP2354B, wired like a Pico 2 (gusmanb's PICO_2 UF2 runs as it is); its decoupling

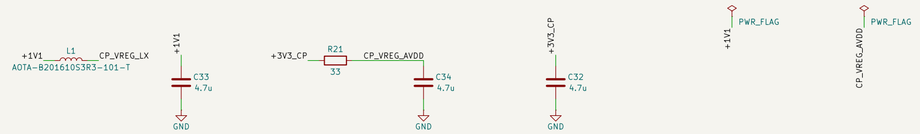




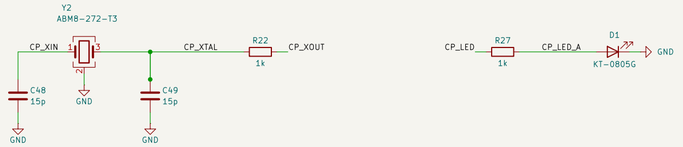
GPIO2-9 = buses A-H (gusmanb CH1-8, 8-bit mode); GPIO0 → GPIO1 bridged (its trigger path).
GPIO14 = the node's mark (TCA9534A P7); GPIO25 = LED; GPIO19…16, 13…10 = pull-up enables A…H (2.2k);
GPIO40-43 = VREF 1-4 / 2 (through 100k); GPIO44 = +5V × 47/147;
  GPIO45 = 3.3 V − (3.3 + 5) × 10/110 from −5 V (2.55 V nominal).
RUN pulled up, BOOTSEL (QSPI_SS) through 1k: both held by the node (TCA9534A), open drain.
Core regulator: Raspberry Pi's Minimal R4 block turned 180°: L1's dot (pad 1) on +1V1, not VREG_LX.
C47 (DVDD pin 32, furthest from the regulator) 4.7 µF: no copper under LX / L1 (In1, In2, B.Cu).

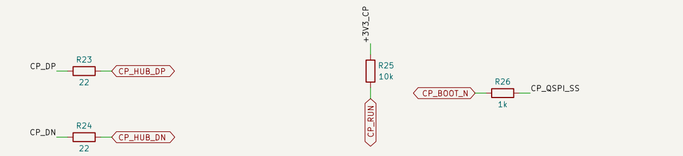
Its 3.3 V from +3V3 through a ferrite, the core regulator as Raspberry Pi lays it out (3.3 µH, 3 × 4.7 µF, 33 Ω)



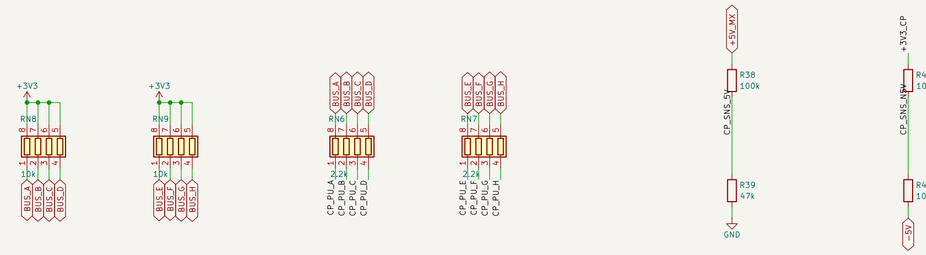


12 MHz crystal (ABM8-272-T3, 15 pF, 1k on XOUT), USB on hub port 5 (22 Ω), RUN, BOOTSEL, LED, SWD





Bus pull-ups: 10k to 3.3 V always (the LSFs' B side), 2.2k switched by the co-processor (GPIO10-13, 16-19); the 15 V rails into the ADC



Generated from electronics/data-port-x4-v2/design.py — edit the Python, then regenerate.
A converted bay's drawer board: four USB-C behind a hub, 32 switchable pins on 8 buses, a logic an
hw-testing-rack
Sheet: /Co-processor/
File: coproc.kicad_sch
Title: DATA ×4 v2 — RP2354B analyzer / co-processor on hub port 5
Size: A2 | Date: | **Rev: 2.0**
KiCad E.D.A. 10.0.6 | Id: 8/8